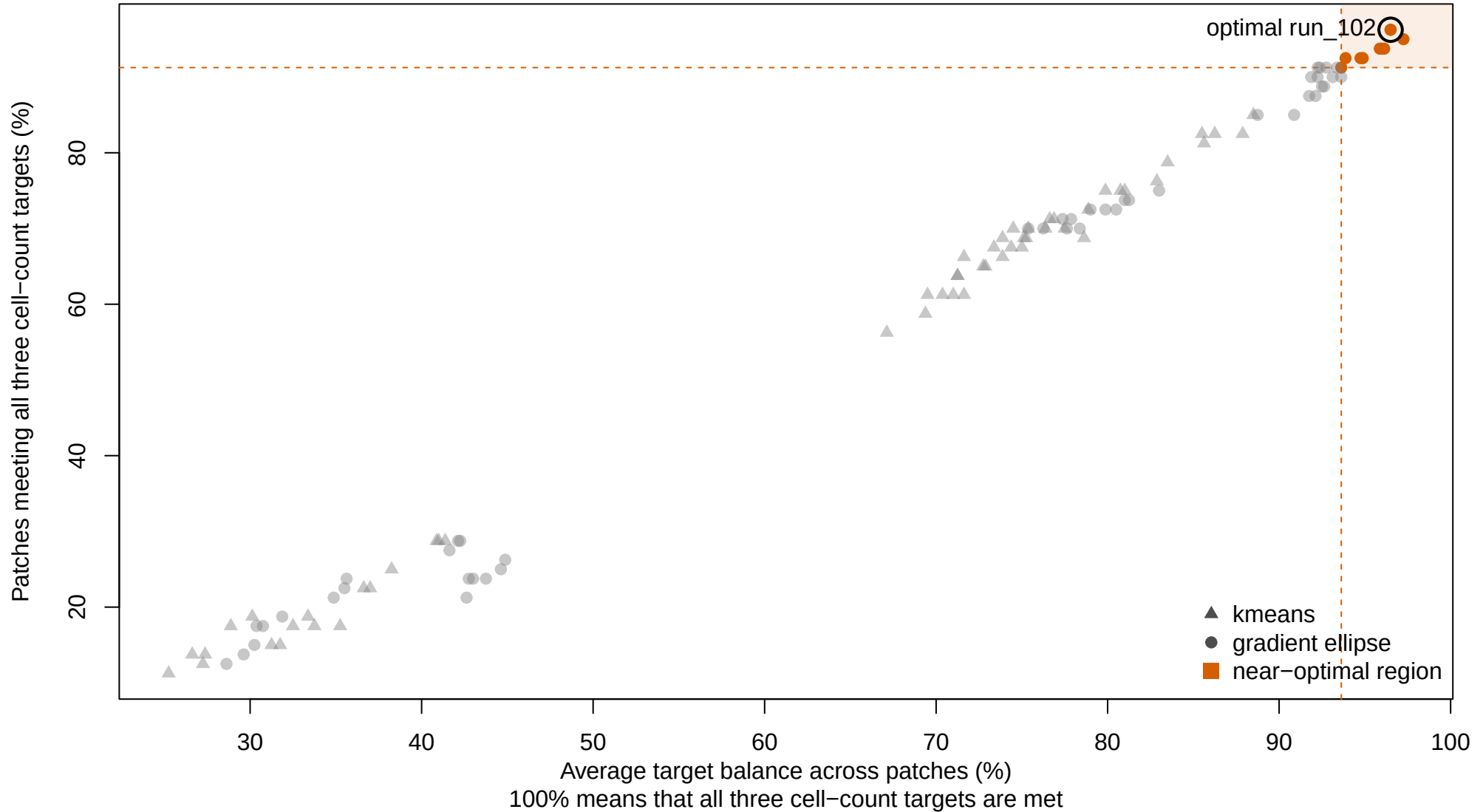


## Biological patch quality across getPatches configurations

Near-optimal configurations rank in the top 10% for both complementary quality criteria

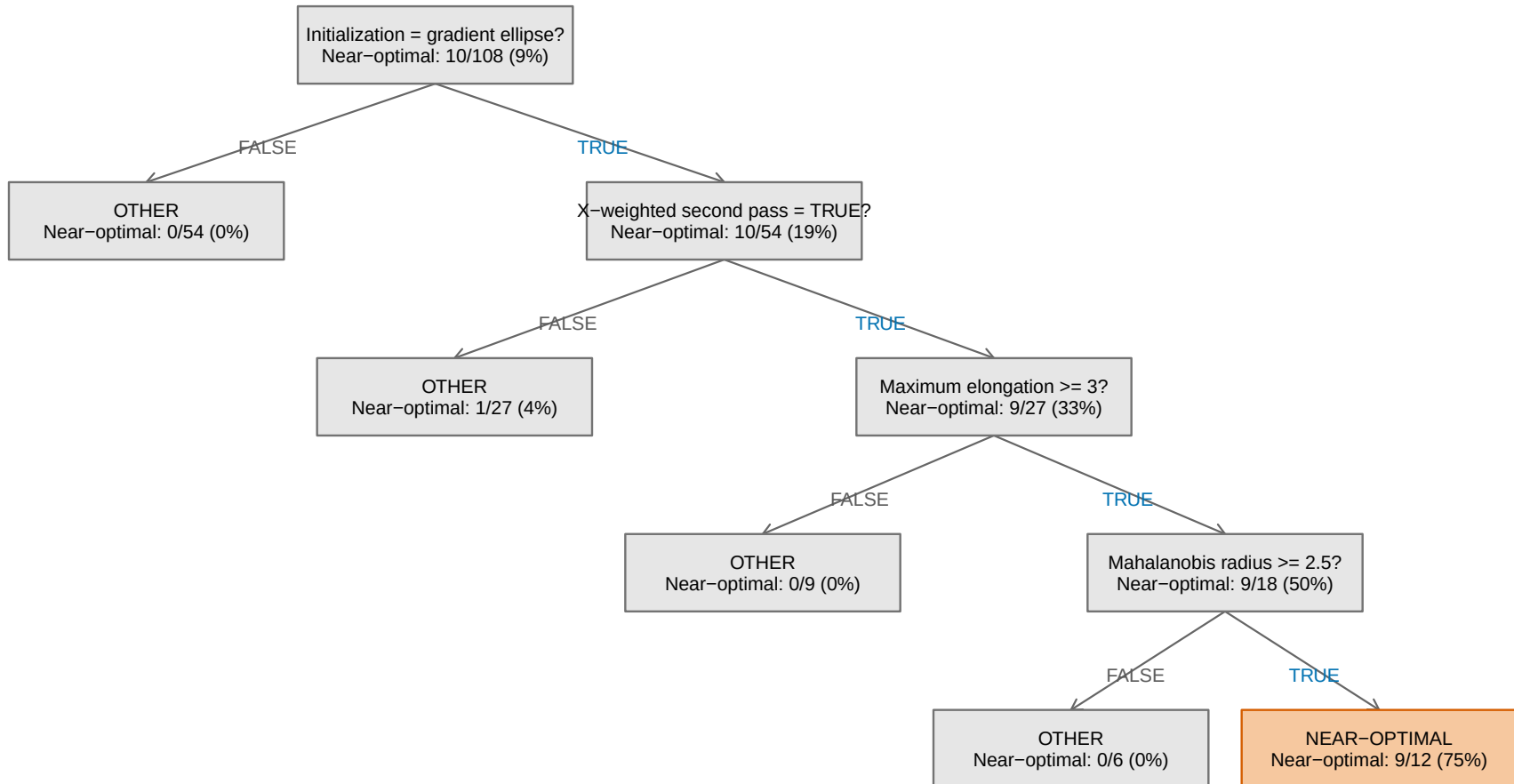
Per-patch targets: Stem cells  $\geq 10$ , Colonocytes  $\geq 10$ , TA  $\geq 10$



CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], n\_iters=10 [default=15],  
init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_radius=NULL [default], mahal\_radius=3 [default], n\_candidates=20 [default], init\_gradient\_k=30 [default],  
init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

## Decision tree: parameter rules identifying near-optimal runs



CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], n\_iters=10 [default=15],  
init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_radius=NULL [default], mahal\_radius=3 [default], n\_candidates=20 [default], init\_gradient\_k=30 [default],  
init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

## Recommended parameter set

Best observed run: run\_102

good patches 96.2% | mean balance 0.965

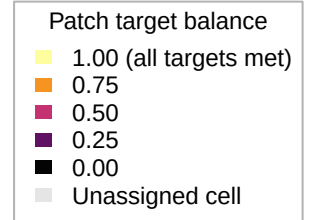
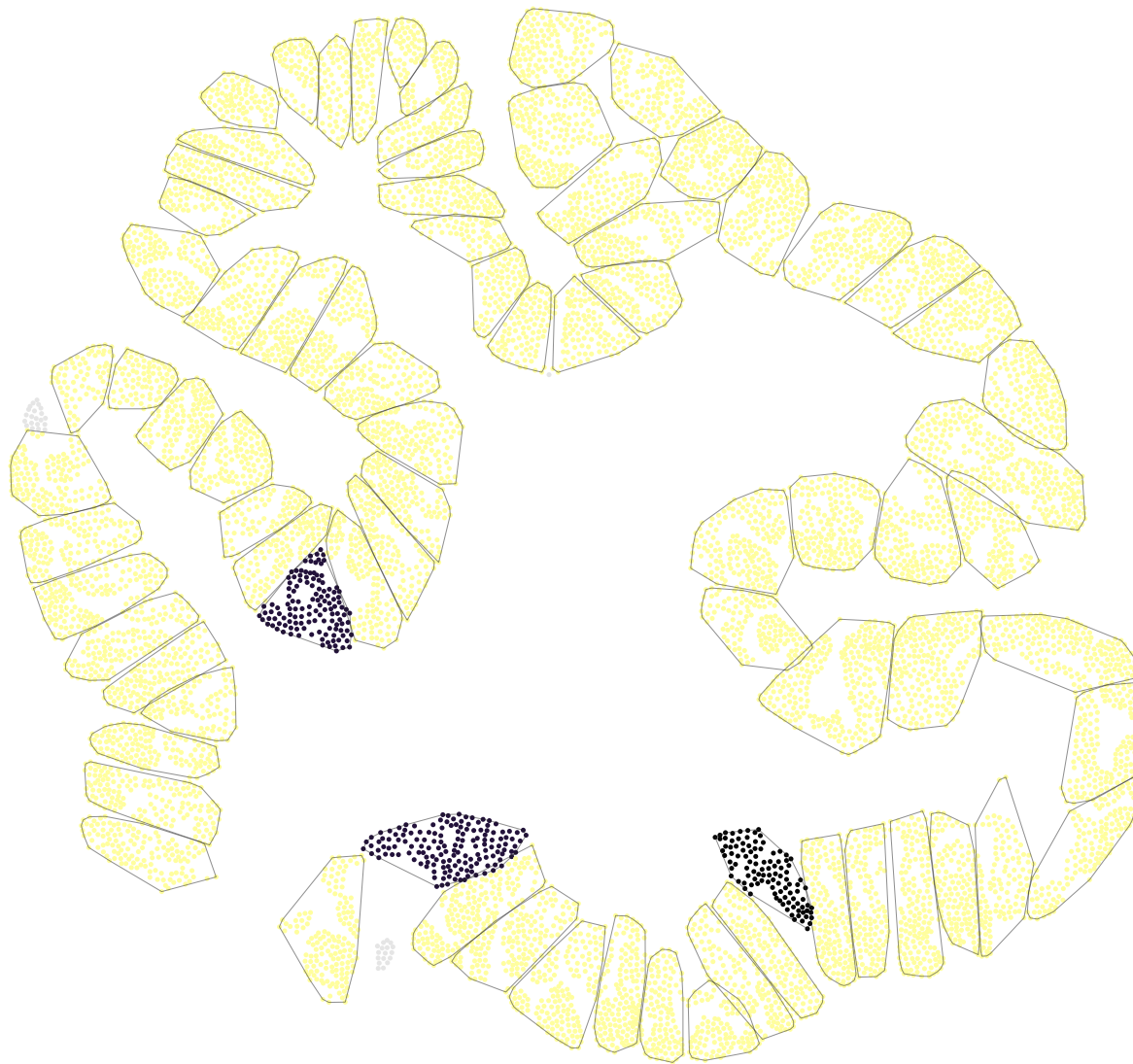
Values equal to getPatches defaults

beta=1 [default], max\_radius=NULL [default], mahal\_radius=3 [default], n\_candidates=20 [default], init\_gradient\_k=30 [default], init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

Values changed from getPatches defaults

npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], n\_iters=10 [default=15], init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

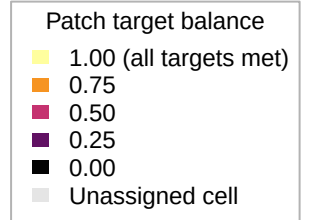
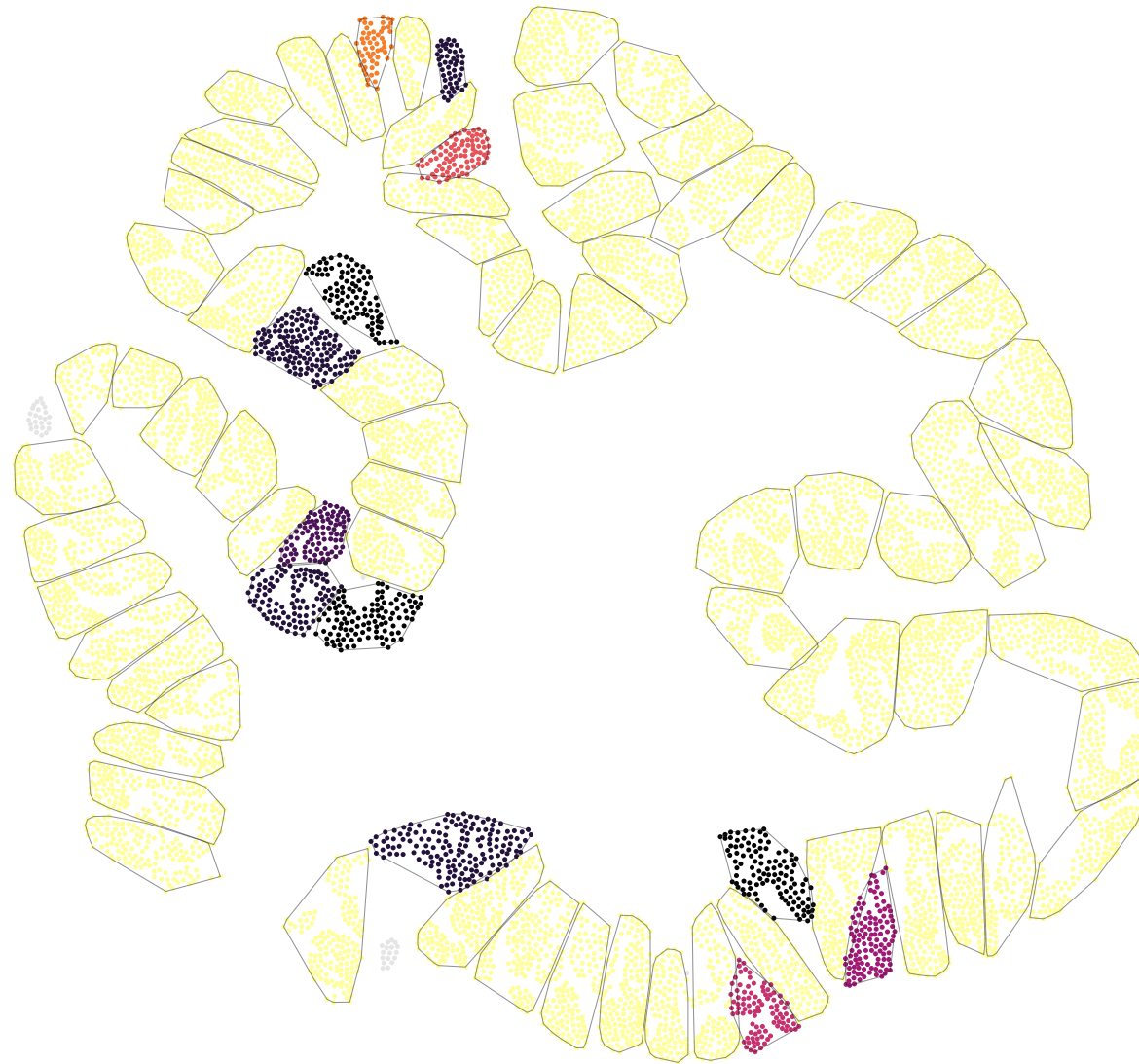
run\_102 | gradient\_ellipse | good 96.2% | balance 0.965



CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], n\_iters=10 [default=15],  
init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_radius=NULL [default], mahal\_radius=3 [default], n\_candidates=20 [default], init\_gradient\_k=30 [default],  
init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

run\_107 | kmeans | good 85.0% | balance 0.885

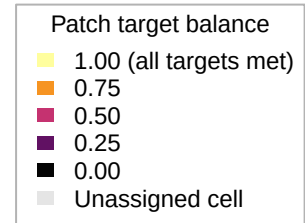
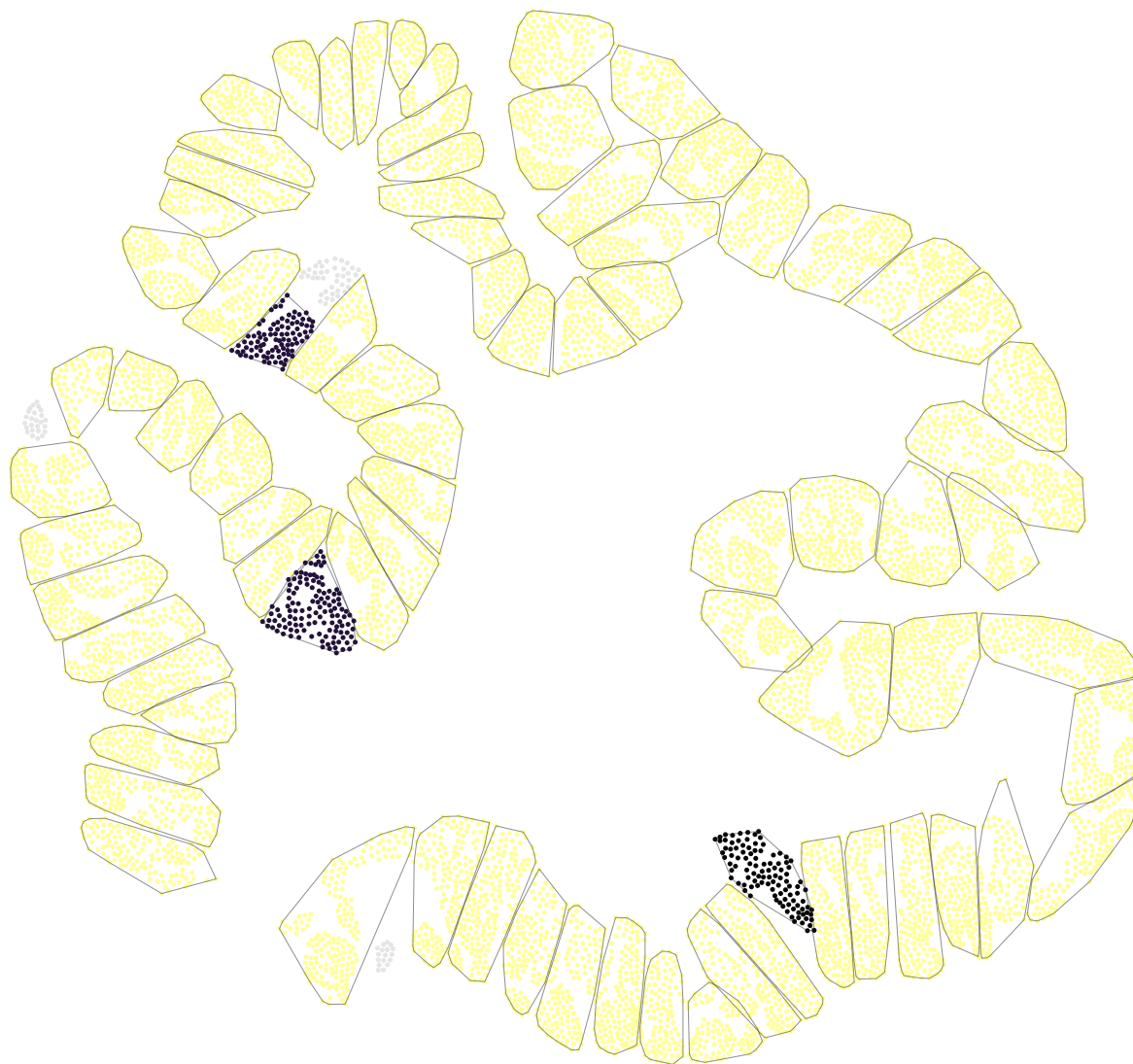


IGNORED: init\_gradient\_k=n/a [default=30], init\_gradient\_elongation=n/a [default=4]

CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], mahal\_radius=4 [default=3], n\_iters=10 [default=15], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_radius=NULL [default], n\_candidates=20 [default], init\_method=kmeans [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

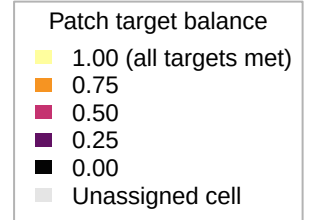
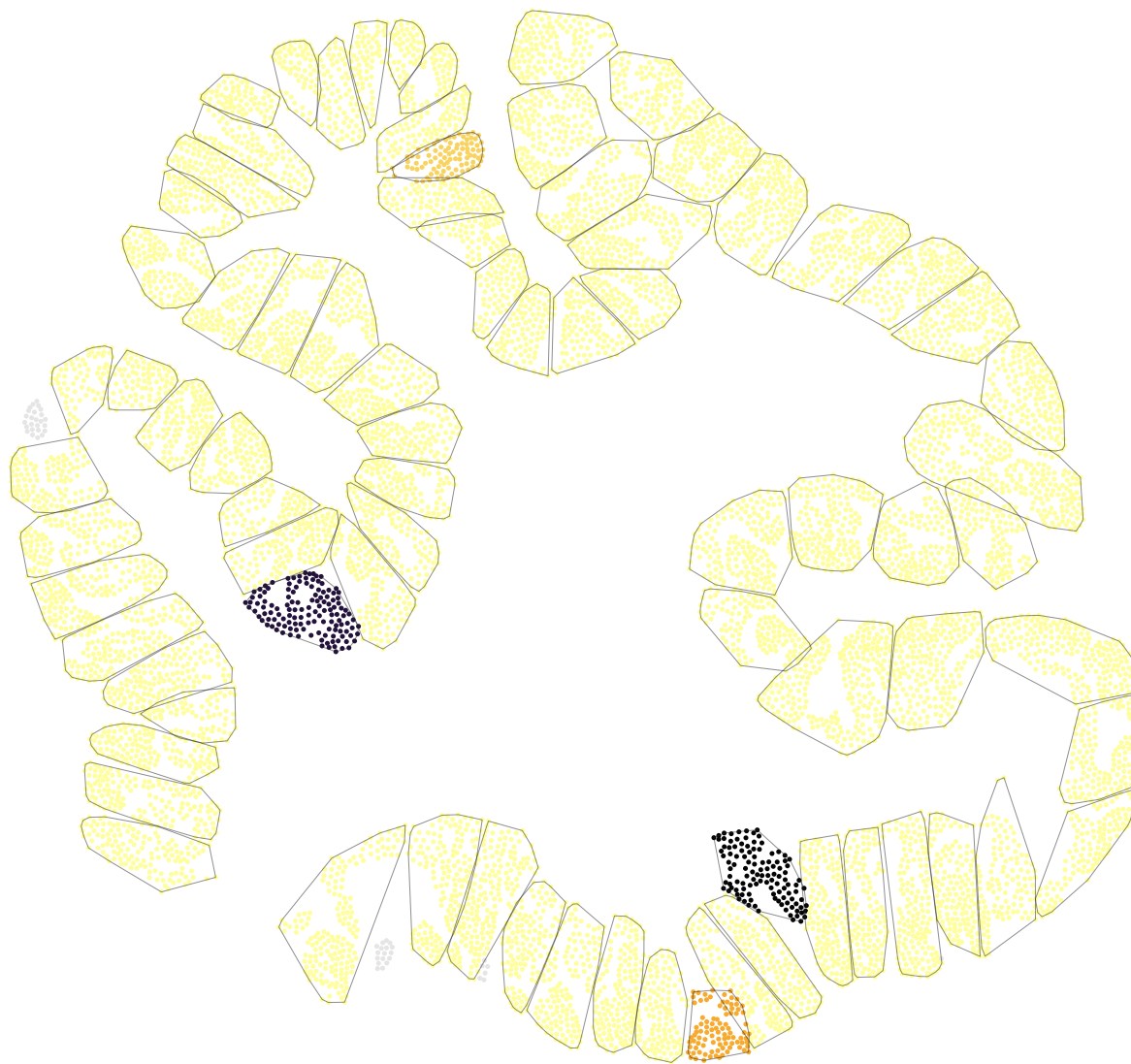
run\_108 | gradient\_ellipse | good 96.2% | balance 0.965



CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], max\_elongation=8 [default=4], mahal\_radius=4 [default=3], n\_iters=10 [default=15], init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_radius=NULL [default], n\_candidates=20 [default], init\_gradient\_k=30 [default], init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]

run\_106 | gradient\_ellipse | good 95.0% | balance 0.973



CHANGED: npatches=80 [required], alpha=0 [default=0.5], hunger\_weight=0.7 [default=0.5], mahal\_radius=4 [default=3], n\_iters=10 [default=15],  
init\_method=gradient\_ellipse [default=kmeans], x\_weighted\_ellipse\_second\_pass=TRUE [default=FALSE]

DEFAULT: beta=1 [default], max\_elongation=4 [default], max\_radius=NULL [default], n\_candidates=20 [default], init\_gradient\_k=30 [default],  
init\_gradient\_elongation=4 [default], x\_ellipse\_gamma=1 [default], x\_ellipse\_wmax=3 [default]